

Scaling the Kitaev/VQE Simulation to Large L

Exact diagonalization, density matrices, matrix product states, and free fermions

Seminar notes

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Abstract

Can the Block-4 noise study be pushed from $L = 4$ toward $L = 16$ (and beyond) on HPC? This note works through the question honestly. The pipeline hides *two* exponential objects with very different escape routes. The **exact-diagonalization reference** is fully removable: the Kitaev Hamiltonian is a free-fermion model, so its ground state and observables follow from a $2L \times 2L$ Bogoliubov–de Gennes (BdG) matrix in $O(L^3)$ time, exact to $L \sim$ hundreds. The **noisy VQE-prepared state** is the hard one: the `EfficientSU2` ansatz is a non-Gaussian, essentially universal circuit, so no classical method escapes exponential cost unconditionally. Matrix product states (MPS) help *when entanglement is low*, and a direct probe of the real ansatz shows the bond dimension χ is bounded *independent of L* at fixed depth ($\chi = 8$ for `reps = 3`, all L). But the same probe exposes a fundamental wall: $\chi \approx 2^{\text{reps}}$ exactly, while the fixed-depth ansatz *loses fidelity* with L ($0.99 \rightarrow 0.02$ over $L = 4 \rightarrow 12$). Keeping the VQE faithful forces `reps` $\sim L$, which sends χ back to $2^{O(L)}$. MPS is cheap precisely where the ansatz fails and expensive precisely where it works — there is no free lunch, and that *is* the NISQ statement.

1 Two exponential objects

The current Block-4 evaluation (`src/block4.py`, plots 8–9) contains two objects that scale exponentially in the number of qubits L :

1. the **ED reference** — `np.linalg.eigh` on the $2^L \times 2^L$ Hamiltonian, used for the ideal $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$ curve, the parity gap, and the fidelity post-selection;
2. the **noisy state** — an exact density matrix ρ of dimension $2^L \times 2^L$ (i.e. 4^L complex entries), produced by `save_density_matrix` on the Aer `density_matrix` method.

The observables themselves are *not* the problem: `qubit_hamiltonian`, `edge_string`, and `parity` are already `SparsePauliOp` objects with $O(L)$ Pauli terms. The blow-up comes entirely from calling `.to_matrix()` on them and from `eigh`. The two objects are addressed separately below.

2 The dense wall

Object 2 sets the practical ceiling. A dense $2^L \times 2^L$ complex 128 matrix costs $16 \cdot 4^L$ bytes, and the pipeline holds several copies (ρ , H , $\hat{P}$, ED eigenvectors, Aer’s internal state):

L	density matrix	+ copies	ED eigh cost
12	0.27 GB	~1 GB	seconds
14	4.3 GB	~20 GB	minutes
15	17 GB	~70 GB	~10 min
16	69 GB	~300 GB	hours

Worse, *every* VQE cost evaluation is a full 4^L density-matrix simulation, run `maxiter` $\times$ `n_starts` times per L . Beyond $L \approx 14$ a single L -point becomes hours to days. Even on a large-memory HPC node the dense path does not reach $L = 16$ in feasible wall-clock time; this is the “dense wall” at $L \approx 15$.

3 Escaping object 1: free fermions

The Kitaev Hamiltonian is quadratic in fermions. After a Jordan–Wigner map the qubit form is

$$H = -\frac{\mu}{2} \sum_j Z_j + \frac{t-\Delta}{2} \sum_j X_j X_{j+1} + \frac{t+\Delta}{2} \sum_j Y_j Y_{j+1},$$

which is a free-fermion (BdG) model. Its ground state, spectrum, correlation matrix, and the parity gap $\Delta_P(L) \sim e^{-L/\xi}$ all follow from diagonalizing a $2L \times 2L$ BdG matrix in $O(L^3)$ time — *no* exponential object. The edge operator $X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ is a product of Majorana operators, so its ground-state expectation is a Pfaffian of a submatrix of the covariance matrix (Wick’s theorem). This is exactly how Majorana splitting is computed for $L \sim 1000$ in the literature. **Conclusion: the ED reference is removable outright**, exact and cheap to $L \sim$ hundreds. This is the content of `src/free_fermion.py`.

4 Escaping object 2: trajectories and MPS

4.1 Quantum-trajectory statevector sampling

Noise (a mixed state) does not require the full density matrix. In the quantum-trajectory picture each shot is a *pure* statevector (2^L , not 4^L) in which the Kraus operators are sampled stochastically; averaging an observable over N_{traj} trajectories reconstructs $\text{Tr}(O\rho)$ up to a statistical error $\propto 1/\sqrt{N_{\text{traj}}}$. Memory drops from 4^L to 2^L (~ 1 MB at $L = 16$), at the honest cost of *error bars* on the noisy curves.

4.2 Matrix product states

A general L -qubit state is a tensor with 2^L amplitudes. An MPS factorizes it into a chain of small tensors, one per site, linked by bond indices of dimension χ :

$$|\psi\rangle = \sum_{\{s\}} A^{s_0} A^{s_1} \cdots A^{s_{L-1}} |s_0 \cdots s_{L-1}\rangle, \quad \text{memory} = O(L \chi^2).$$

Across any cut, χ equals the number of nonzero Schmidt coefficients — a direct measure of *entanglement* across that cut. A gapped 1D ground state obeys an area law, so its entanglement is bounded independent of L ; then χ is constant and MPS is *linear* in L . A critical or volume-law state needs $\chi \sim 2^{L/2}$ and MPS is exponential again. Aer provides `method='matrix_product_state'` and supports noise through trajectory sampling; the empirical test confirms it runs and exposes the per-bond χ .

4.3 Truncation error, and why the risk is controllable

A two-qubit gate can double χ across its bond. To stay cheap one performs an SVD there and keeps only the largest $\chi_{\max}$ Schmidt values. The discarded weight — the sum of squared discarded Schmidt coefficients — *is* the truncation error: the fraction of the state’s norm thrown away. If nothing is discarded the MPS is *exact*. If $\chi_{\max}$ is set below what the state needs, the simulated state becomes a nearby but different state and $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$ acquires a systematic bias. Crucially this is a knob with a readout: Aer reports the discarded weight, and the convergence check “run at $\chi_{\max}$ and $2\chi_{\max}$; if the observable does not move, the result is effectively exact” makes the error visible and quantitative rather than hidden. A truncated number never silently corrupts a result as long as the discarded weight is logged.

5 The χ -probe: what the real ansatz actually does

We measured the *exact* (uncapped) max bond dimension of the real `EfficientSU2(ry, linear, reps)` state directly.

χ is flat in L at fixed depth. Worst-of-random- θ max χ (reps = 3):

L	6	10	14	18	22
χ	8	8	8	8	8
$2^{\lfloor L/2 \rfloor}$ (vol. law)	8	32	128	512	2048

A reps = r linear ansatz has a light cone of width $\sim r$, so entanglement across any cut is bounded *independent of* L . At reps = 3, χ pins at 8 for all L , far below the volume-law ceiling, and noise does not inflate it. **The truncation risk essentially does not exist for a shallow ansatz:** cap $\chi_{\max} = 16$ and the simulation is exact to $L = 100$.

χ grows exponentially in depth. At fixed $L = 14$:

reps	1	2	3	4	5	6	8
χ	2	4	8	16	32	64	128

i.e. $\chi \approx 2^{\text{reps}}$ (until it saturates the $2^{L/2}$ ceiling).

The fixed-depth ansatz loses the state. VQE fidelity to the true ground state at reps = 3:

L	4	6	8	10	12
fidelity	0.99	0.50	0.25	0.10	0.02

A fixed-depth hardware-efficient ansatz simply cannot represent the topological ground state as L grows; keeping the fidelity up requires reps $\sim L$.

6 The wall is fundamental, not a coding artifact

Combining the three tables: the regime where MPS is cheap (small reps, constant χ) is exactly the regime where the VQE *fails* (fidelity $\rightarrow 0$); and the regime where the VQE *works* (reps $\sim L$) is exactly where $\chi \sim 2^{\text{reps}} \sim 2^{O(L)}$ blows MPS back up. A state that genuinely needs a deep circuit to

prepare is genuinely expensive to simulate classically — MPS buys a smaller exponential base (2^{reps} vs 4^L), a few extra qubits, not unbounded L . This is the honest simulation-cost face of “NISQ does not inherit topological protection.”

The one unconditionally polynomial regime. The true Kitaev ground state is Gaussian and area-law: it *has* a small- χ MPS. The `ry` ansatz is merely an inefficient, non-Gaussian parametrization of it. A **matchgate/Gaussian ansatz** (only R_z -type single-qubit gates + nearest-neighbor matchgate entanglers) prepares the same state at $O(L)$ depth with bounded χ , and under *fermion-Gaussian* noise (amplitude damping/loss is Gaussian for fermions; depolarizing is not) the entire simulation reduces to covariance-matrix algebra, $O(L^3)$, exact, $L \rightarrow$ thousands. The catch: this answers a cleaner but *different* question — it is no longer the generic hardware-efficient NISQ story.

7 Recommended architecture

piece	scalable replacement	exact?
ideal $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$, gap	free-fermion BdG (L^3)	exact, $L \rightarrow$ hundreds
noisy VQE $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$	MPS + noise trajectories ($L\chi^2$)	exact if χ bounded
unbounded L	matchgate ansatz + Gaussian noise (L^3)	exact, different model

The concrete plan: build `src/free_fermion.py` (an unconditional win) and pair it with a *fixed-reps* MPS sweep that makes the ansatz’s expressibility collapse visible against the exact ideal curve. That is honest, cheap, reaches large L , and turns the wall itself into the scientific point. Validation anchor: at $L \leq 12$, where the exact density-matrix path still runs, cross-check the MPS-trajectory and free-fermion values against the exact numbers before extrapolating.

Backends for the noisy curve. A device-calibrated sweep needs a snapshot whose coupling map embeds a length- L line so the linear-entanglement `cx` pairs sit on real edges with no SWAP inflation. Named 5-qubit snapshots (`fake_manila`, etc.) cap at $L = 5$; `FakeGuadalupeV2` is 16 qubits but its heavy-hex graph only contains a 13-node line. The clean choices are the 27-qubit heavy-hex snapshots (`FakeCairoV2`/`FakeKolkataV2`/`FakeHanoiV2`, 21-node line) or a 20-qubit device (18-node line).

Reproduce. χ -probe: `python scratch/chi_probe.py` (measures χ vs L and vs reps against the volume-law ceiling). Free-fermion reference: `src/free_fermion.py` (BdG ground state, edge string, parity gap; validated against `eigh` at $L \leq 12$).